

NETWORK SCANNING AND INFORMATION GATHERING USING NMAP

Internship Project Report

Prepared By

Debraj Roy

Tools Used

- Kali Linux 2025.1
 - Nmap 7.95
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1. Introduction

Network scanning and information gathering are fundamental activities in cybersecurity assessments. Before performing any vulnerability analysis or penetration testing, security professionals must identify active devices, open ports, available services, and operating systems within a network.

This project demonstrates the use of Nmap, a widely used network scanning tool, to perform reconnaissance on a controlled local network environment.

2. Objectives

The objectives of this project were:

- Identify active hosts on the local network.
 - Discover open ports on target systems.
 - Detect running services and service versions.
 - Perform operating system fingerprinting.
 - Analyze network exposure from a security perspective.
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3. Tools and Environment

Operating System

Kali Linux 2025.1

Scanning Tool

Nmap 7.95

Network Environment

Private home network used for authorized testing.

4. Methodology

Step 1: Identify Local IP Address

The IP address of the scanning system was identified using:

```
ip addr
```

The scanning machine was assigned an address within the 192.168.0.0/24 network range.

Step 2: Host Discovery

A ping sweep was conducted to identify active devices on the network.

Command Used:

```
nmap -sn 192.168.0.0/24
```

This scan identified active hosts within the network segment.

Step 3: Service Enumeration

A detailed scan was performed against the selected target host.

Command Used:

```
nmap -sV 192.168.0.102
```

The scan detected open ports and service versions running on the target system.

Step 4: Operating System Detection

Operating system fingerprinting was conducted using:

```
sudo nmap -O 192.168.0.102
```

This enabled identification of the probable operating system running on the target host.

5. Results

Host Discovery Results

Active hosts identified:

- 192.168.0.1 (Network Router)
 - 192.168.0.102 (Target System)
 - 192.168.0.108 (Scanning System)
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Service Detection Results

Port	State	Service	Version
80	Open	HTTP	Microsoft IIS 10.0

The presence of Microsoft IIS indicates that the target system was running a web service accessible over HTTP.

Operating System Detection Results

Nmap OS fingerprinting suggested:

- Microsoft Windows 11 (Highest Probability)
- Microsoft Windows 10 (Possible Match)

The target host was determined to be a Windows-based system.

6. Security Analysis

The target host exposed an HTTP service on Port 80.

Potential security considerations include:

- Web server misconfiguration
- Outdated software versions
- Unnecessary service exposure
- Weak access controls

Regular vulnerability assessments and patch management should be implemented to reduce risk.

7. Conclusion

This project successfully demonstrated network reconnaissance techniques using Nmap.

The assessment included host discovery, service enumeration, and operating system detection. The project provided practical experience in information gathering, which serves as the foundation for subsequent cybersecurity assessments and penetration testing activities.

The skills learned during this project are directly applicable to real-world security operations and vulnerability assessment processes.

8. References

- Nmap Official Documentation
- Kali Linux Documentation
- Network Security Best Practices